Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
E		2602872

Solar Sculptors: An Optical-Thermal Coupled Framework for Passive Solar Shading Design

Summary

Passive solar shading must balance three competing needs: reducing unwanted summer solar gains, preserving useful winter gains, and controlling glare without over-darkening interiors. Common sizing rules based on a few static sun angles cannot represent hourly sun paths, strong east west asymmetry, or the time delay between solar input and HVAC demand caused by **thermal inertia**. To address 2026 ICM Problem E, we develop a **time-resolved solar shading thermal framework** for retrofitting Academic Hall North at two contrasting campuses (**Sungrove**: cooling-dominated; **Borealis**: heating-dominated), extracting transferable design rules, and evaluating a Student Union prototype using the same performance metrics.

The model follows a physical chain. Hourly solar geometry provides facade-specific incidence angles. Shading performance is computed as a time-dependent beam-blocking ratio from window shadow overlap via the **Sutherland–Hodgman algorithm** (polygon clipping), applicable to overhangs, fins, and composite screens. **Window solar heat gain** is then formed by combining direct, diffuse, and ground-reflected components with glazing solar admission. Indoor response is computed with a reduced-order **resistance–capacitance (RC) model** for rapid screening, and key cases are checked with a **transient field model** to confirm window-adjacent hotspots, stratification, and surface temperatures used in operative temperature and radiant asymmetry. Design variables (L_{overhang}, L_{fin}, τ_{eff}, C_{eff}) are selected by first screening **non-dominated candidates** and then choosing a single implementable design by **distance-to-ideal ranking** under energy, comfort, and daylight criteria.

In Sungrove, baseline conditions (double-pane glazing, no shading) show severe overheating: with a peak outdoor temperature of **38** ℃, the indoor peak reaches **48.4** ℃, and summer-solstice window solar heat gain totals **576** kWh. The recommended retrofit uses $L_{\text{overhang}} = \mathbf{1.0}\,\text{m}$, $L_{\text{fin}} = \mathbf{0.8}\,\text{m}$, $\tau_{\text{eff}} = \mathbf{0.40}$, and $C_{\text{eff}} = 1.5\times$. It reduces east west daily heat gain from **120** kWh to **35** kWh, lowers total daily window solar heat gain to **212** kWh (**63.1%** reduction), and decreases mean indoor temperature from **36.1** ℃ to **26.5** ℃. The cooling-load peak occurs about **3.0 h** after the peak in window solar heat gain, consistent with **thermal lag**. Vertical fins target low-angle east and west sun that drives glare and morning and afternoon gains.

In Borealis, winter solar altitude can fall to **1.87** ° at noon, limiting useful solar input to a short daily window. The recommended strategy increases glazing **solar admission** (triple-pane units, SHGC=**0.65**) and strengthens **thermal inertia** to bridge the short solar window, store daytime gains, and release them at night. The coupled retrofit yields a thermal lag of about **2.8 h** and reduces nighttime peak heating demand by nearly **40%**, while improving envelope surface temperatures and reducing cold radiant asymmetry near windows.

Model checks support numerical reliability: solar position matches NREL SPA with MAE **0.34** ° in elevation and **0.51** ° in azimuth; a grid-independence check gives mean-temperature deviation **0.3%** for an 80×40 grid; and the 14 h energy balance closes within $\varepsilon_E = \mathbf{1.1\%}$. Overall, cooling-dominated climates prioritize **beam rejection** (geometry first, then lower τ_{eff}), while heating-dominated climates prioritize **solar admission** paired with larger effective thermal mass, with daylight and comfort as binding checks.

Keywords:

passive solar shading; solar geometry; window solar heat gain; transient heat transfer; thermal mass; comfort and daylight; climate transferability

Contents

1 Introduction

1.1 Problem Background

The accelerating impacts of climate change have imposed unprecedented pressure on the built environment to decarbonize, with buildings responsible for nearly 40% of global emissions. Passive shading offers a critical low-carbon pathway to regulate solar heat gain and reduce reliance on mechanical systems through optimized geometric and material design.

However, conventional shading design methods are typically anchored to static solar geometry at solstice noon, neglecting the coupled influences of diurnal solar motion, seasonal climatic variability, and building thermal inertia. This simplification is particularly consequential when applied across fundamentally distinct climatic regimes, such as cooling-dominated low-latitude and heating-dominated high-latitude contexts.

To achieve robust, future-proof performance, next-generation shading strategies must move beyond static snapshots and incorporate time-resolved solar modelling, climate-specific optimization, and transient thermal response mechanisms. Addressing these dual challenges requires a first-principles, optical-thermal coupled framework capable of systematically optimizing shading performance across contrasting environmental contexts.

1.2 Problem Restatement

Commissioned by COMAP, we provide model-based feasibility analyses and design proposals for passive solar shading strategies at Sungrove and Borealis Universities. The core objective is to balance thermal comfort, energy efficiency, and visual performance through mathematical modelling. We decompose the problem into the following five core tasks:

Task 1: Retrofit Analysis for Sungrove University (Cooling-Dominated).
Objective: Design retrofit strategies for the existing "Academic Hall North" to reduce cooling loads and resolve glare issues.
Core Requirement: The design must align with the "Net-Zero Cooling by 2040" initiative. This involves developing a model to evaluate the effectiveness of passive strategies (e.g., overhangs, high-performance glazing) in rejecting solar heat throughout the academic year, transcending traditional static calculations at noon.

Task 2: Extended Application to Borealis University (Heating-Dominated).
Objective: Adapt the retrofit model for Borealis University, which shares the same building footprint but faces a heating-dominated climate.
Core Focus: The model must explicitly quantify the role of **Thermal Mass** in capturing and storing solar energy. The challenge lies in maximizing winter heat gain while preventing overheating during long summer daylight hours, requiring a distinct strategy from Sungrove.

Task 3: Global Generalizability and Adaptive Expansion.
Objective: Demonstrate model robustness by extending the analysis beyond these two specific locations.
Core Requirement: Discuss design considerations for other locations with similar latitudes but potentially differing microclimates or heating/cooling demands. This requires abstracting

model parameters to form a universal design framework adaptable to global contexts.

Task 4: Prototype Design for the New Student Union
Objective: Design a comprehensive passive shading strategy for a new Student Union building at either Sungrove or Borealis University.
Core Vision: Create a "Prototype for the Future" that prioritizes passive cooling over mechanical systems. The design should integrate innovative features (e.g., adaptive skins, non-standard geometries) and ensure resilience against future climate projections.

Task 5: Strategic Recommendation Letter to University Leadership.
Objective: Synthesize technical findings into a 1-2 page letter addressed to the leadership of the selected university (Sungrove or Borealis).
Content: Provide actionable steps for implementing passive solar shading in both retrofit and new-build projects, supported by quantitative evidence derived from the modeling results.

1.3 Our Work

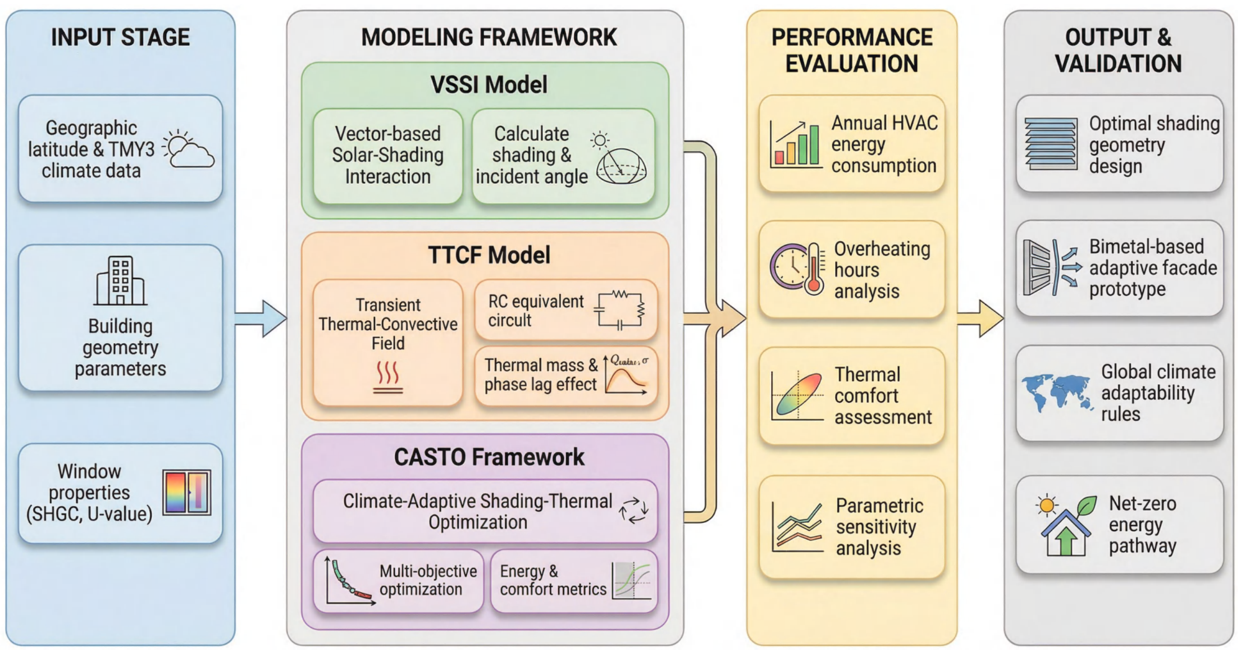

Figure 1: Overview of our work.

2 Preliminaries

2.1 Assumptions and Justifications

Assumption 1 (1D Transient Conduction): *Heat transfer through the building envelope is modeled as dominant one-dimensional conduction along the thickness direction, while lateral thermal bridges and edge effects are treated as second-order correction terms.*

Justification 1: Given that the surface area scale of large-scale building walls is far greater than the thickness scale, scale analysis indicates that heat flow perpendicular to the surface dominates the "absorption-storage-release" time-lag process in Borealis University's winter.

Assumption 2 (Optical-Thermal Decoupling): *External shading devices are treated as pure geometric optical operators that regulate incident radiation flux solely through projection, without participating in the system thermal balance as heat capacity nodes.*

Justification 2: Given the order-of-magnitude difference between optical and thermal diffusion speeds, secondary radiation from shading devices is primarily dissipated outdoors, making its indoor cooling load contribution negligible.

Assumption 3 (Hierarchical Thermal Modeling): *In the parameter screening phase, indoor air is assumed to be fully mixed with uniform temperature everywhere; in the detailed evaluation phase, the three-dimensional temperature field is restored to capture window-side hotspots and vertical thermal stratification.*

Justification 3: This hierarchical modeling strategy enables us to scan thousands of combinations of shading depths and glazing parameters within a reasonable time, while ensuring the final recommended solution withstands rigorous verification of local comfort for window-side seating.

Assumption 4 (Isotropic Diffuse Radiation): *Beam radiation is calculated precisely using vector projection, while diffuse and ground-reflected radiation are assumed to be isotropic.*

Justification 4: Under the intense sunlight of Sungrove, noon beam irradiance can reach 1000 W/m^2, whereas diffuse irradiance is only around 200 W/m^2; simplifying the secondary anisotropic diffuse field significantly improves optimization efficiency.

Assumption 5 (Quasi-Static Boundary Conditions): *Within each discrete simulation time step, the solar position vector and meteorological boundary conditions (temperature, irradiance) are maintained as quasi-static.*

Justification 5: The hourly time step satisfies the *Nyquist* sampling theorem for capturing diurnal thermal cycles while filtering high-frequency meteorological noise (eg. passing clouds) that has a negligible impact on the thermal response of high-thermal-mass buildings.

2.2 Data Description

Table 1: Data sources.

Source	**Description**
Problem E Statement	60×24 m footprint, 45% South WWR; Sungrove 20°N, Borealis 65°N. Defines geometric domain and climatic targets.
NREL TMY3 [1]	Honolulu (911820), Fairbanks (702610); hourly GHI, DNI, DHI, T. Boundary conditions for transient simulation.
LBNL IGDB v102.0 [2]	Spectral $\tau(\lambda)$, $\rho(\lambda)$ for triple-pane/low-E glass; used for SHGC_{eff} in the *VSSI* model.
ASHRAE Handbook [3]	1.1 met, clothing (clo); comfort band 20–26°C. Objectives and constraints for optimization.

2.3 Notations

Table 2: Nomenclature

Symbol	Definition	Unit
ϕ	Geographical Latitude	$\circ$
δ	Solar Declination Angle	$\circ$
ω	True Solar Hour Angle (0 at solar noon)	$\circ$
α_s	Solar Altitude Angle	$\circ$
γ_s	Solar Azimuth Angle (Clockwise from North)	$\circ$
θ_i	Solar Incidence Angle (Relative to window normal)	$\circ$
$F_s(t)$	Instantaneous Shading Coefficient (Shading Fraction)	$[0, 1]$
$\mathcal{G}$	Geometric Projection Shading Operator	$[0, 1]$
$\mathcal{M}$	Material Transmission Operator (Effective SHGC)	$[0, 1]$
$\dot{Q}_{\text{solar,win}}$	Instantaneous Total Solar Heat Gain Power of Window	W
$T(\mathbf{x}, t)$	Indoor 3D Transient Temperature Field	°C
k_{eff}	Effective Turbulent Thermal Diffusivity	$\text{W}/(\text{m} \cdot \text{K})$
R_{eq}	Room Equivalent Total Thermal Resistance	K/W
C_z	Room Equivalent Total Thermal Capacitance	J/K
$\dot{Q}_{\text{HVAC}}$	Ideal HVAC Heat Load (+ Heating / - Cooling)	W
t_{lag}	Thermal Wave Transmission Lag Time	h
$\mathbf{x}$	Optimization Decision Vector	—
L_{overhang}	Horizontal Overhang Depth	m
L_{fin}	Vertical Fin Depth	m
τ_{eff}	Window System Equivalent Transmittance	$[0, 1]$
C_{eff}	Thermal Mass Enhancement Multiplier	$[1, 3.0]$
T_{op}	Indoor Operative Temperature (Combined Air and Radiant)	°C
η_{comfort}	Annual Thermal Comfort Coverage Rate	%
DF_{proxy}	Daylighting Proxy Metric	%
J_{load}	Annual Cumulative Energy Load	kWh

3 Model Development

This chapter constructs a mathematical framework coupling "Solar Geometry – Dynamic Shading – Transient Thermal Response." The modeling logic follows the energy flow path: first, determining hourly solar positions and calculating shading truncation via vector projection; second, converting environmental radiation into indoor heat sources through decomposition and window transmission; finally, establishing transient energy balance equations to solve the spatiotemporal evolution of the indoor temperature field.

3.1 Solar Geometry–Shading Model

Shading effectiveness is fundamentally determined by the solar trajectory. To avoid limiting the evaluation to static snapshots (e.g., noon on the solstice), this section establishes a time-resolved solar position and incidence geometry. On this basis, we define the shading factor $F_s(t)$ to characterize the attenuation of the direct solar component by shading elements.

3.1.1 3D Solar Geometry

In the local building coordinate system (x East, y North, z Up), the solar position is represented by the solar altitude angle $\alpha_s(t)$ and the solar azimuth angle $\gamma_s(t)$. To achieve high-precision hourly simulation, we adopt the solar position algorithm of Reda and Andreas [4]. First, the solar declination δ is given by the day of the year n:

$$\delta = 23.45 \sin\left(\frac{360}{365}(n-81)\right) \tag{1}$$

(converted to radians for calculation). Combining the geographic latitude ϕ and the true solar hour angle $\omega(t)$, the solar altitude α_s satisfies:

$$\sin\alpha_s = \sin\phi\sin\delta + \cos\phi\cos\delta\cos\omega \tag{2}$$

When $\alpha_s < 0$, the sun is considered below the horizon, and the direct component is taken as 0. To unify the geometric relationship with windows of arbitrary orientation, we denote the solar direction unit vector as:

$$\mathbf{S} = [\cos\alpha_s\sin\gamma_s,\ \cos\alpha_s\cos\gamma_s,\ \sin\alpha_s]^T \tag{3}$$

The normal vector of an arbitrary window can be written as:

$$\mathbf{N} = [\sin\beta\sin\psi,\ \sin\beta\cos\psi,\ \cos\beta]^T \tag{4}$$

where ψ is the window azimuth and β is the window tilt angle (for vertical windows, $\beta = 90°$). Thus, the solar incidence angle is concisely given by the dot product:

$$\theta_i(t) = \arccos\Big(\mathbf{S}(t)\cdot\mathbf{N}\Big) \tag{5}$$

When $\alpha_s < 0$ (night) or $\theta_i > 90°$ (self-shadowing), the contribution of direct radiation is zero. This vector expression constitutes the geometric cornerstone for all subsequent shading calculations, ensuring the model can naturally handle arbitrary conditions from vertical windows to skylights, and from south-facing to northwest-facing orientations.

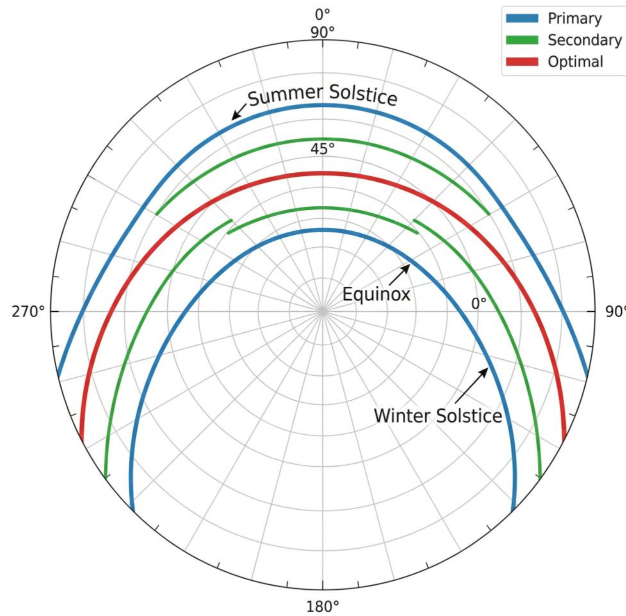

Figure 2: Solar polar path diagram. The radial direction represents the azimuth γ_s, and the concentric circles represent the altitude α_s. The blue lines indicate the paths for the representative days of Summer Solstice, Equinox, and Winter Solstice.

3.1.2 VSSI Model

In architecture, while facade shading elements vary in form (Kirimtat et al. [5]), they can be optically classified into three categories (Figure 3):

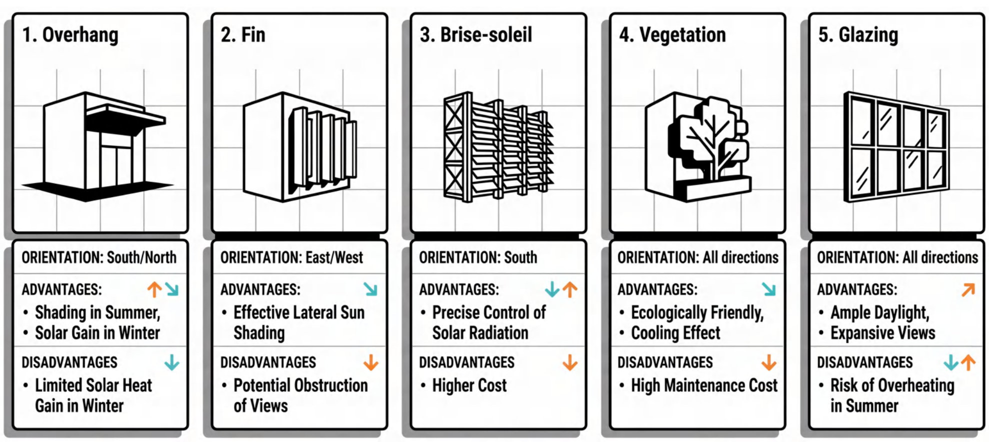

Figure 3: Topological classification of facade shading elements.

To mathematically unify these diverse forms, we propose the **Vector Solar-Shading Interaction (*VSSI*)** model. This model decouples the shading effect into two operators: the macroscopic **Geometric Operator** $\mathcal{G}(\vec{S}(t), \mathbf{p}_{\text{geo}})$, which describes the proportion of direct light blocked under the solar vector $\vec{S}(t)$ (corresponding to the hourly shading factor $F_s(t)$);

and the microscopic **Material Operator** $\mathcal{M}(t, \mathbf{p}_{\text{mat}})$, which describes the proportion of radiation converted into indoor heat gain after penetrating the medium (i.e., the effective Solar Heat Gain Coefficient, $\text{SHGC}_{\text{eff}}(t)$).

Specifically, for the geometric operator $\mathcal{G}$, we define the shading factor $F_s(t)$ as the area ratio between the shading projection $S_{shadow}(t)$ and the window opening W:

$$F_s(t) \equiv \mathcal{G} = \frac{A(W \cap S_{shadow}(t))}{A(W)} \tag{6}$$

In general cases (e.g., polygonal windows or complex shading), the overlapping area $A(W \cap S_{shadow})$ can be precisely solved via the **Polygon Clipping Algorithm**. As Figure 4, using the *Sutherland–Hodgman* algorithm [6], the exact illuminated area is obtained by iteratively cutting the window polygon with a series of half-planes defined by the shadow boundaries.

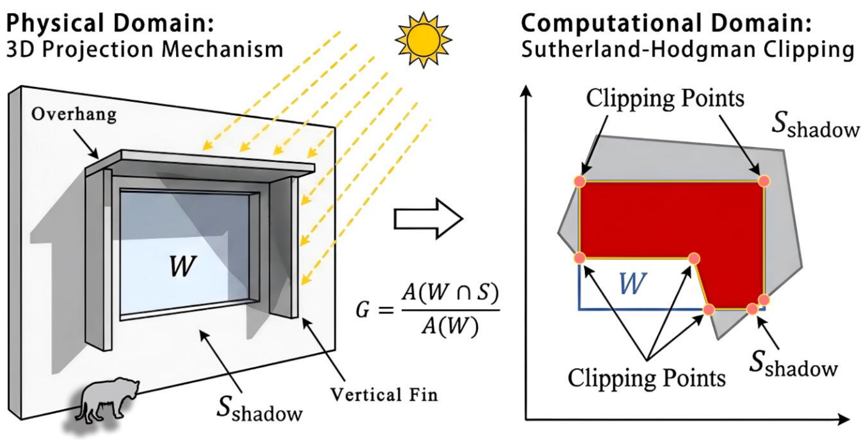

Figure 4: Geometric occlusion (projection–clipping); much faster than pixel-counting.

Based on this decoupled framework, the effective direct heat gain $\dot{Q}_{\text{beam}}(t)$ entering the room at any moment can be directly expressed as the product of the two operators:

$$\dot{Q}_{\text{beam}}(t) = A_{\text{win}} \cdot \underbrace{\cos\theta_i \cdot I_{\text{beam}}(t)}_{\text{Incident Beam}} \cdot \underbrace{\left[1 - \mathcal{G}(\mathbf{S}(t), \mathbf{p}_{\text{geo}})\right]}_{\text{Geometry: Shading}} \cdot \underbrace{\mathcal{M}(t, \mathbf{p}_{\text{mat}})}_{\text{Material: Transmission}} \tag{7}$$

To ensure complete energy accounting, the total solar heat gain of the window must also include diffuse and ground-reflected components. Based on the ***Liu & Jordan* isotropic sky model** [7], the diffuse radiation on a tilted surface is related to the sky view factor, while the ground-reflected radiation depends on the ground albedo and tilt angle. Instantiating $\mathcal{M}$ as the SHGC, the final instantaneous solar thermal power entering the room, $\dot{Q}_{\text{solar,win}}(t)$, is:

$$\dot{Q}_{\text{solar,win}}(t) = \dot{Q}_{\text{beam}}(t) + A_{\text{win}} \cdot \mathcal{M} \cdot \left[\underbrace{I_{\text{diffuse}}(t)\frac{1+\cos\beta}{2}}_{\text{Diffuse}} + \underbrace{\rho_g I_{\text{GHI}}(t)\frac{1-\cos\beta}{2}}_{\text{Reflected}} \right] \tag{8}$$

At this point, we have completed the full mapping from "Solar Position" to "Indoor Heat Source." This heat source term $\dot{Q}_{\text{solar,win}}(t)$ will serve as the core driving force input into the indoor thermal response model in the next section.

3.2 Indoor Transient Thermal Response

The impact of shading systems on building performance is not limited to the reduction of total energy but profoundly alters the spatiotemporal distribution of heat. The solar radiation $\dot{Q}_{\text{solar,win}}(t)$ calculated in Section 3.1 does not act uniformly on the interior; instead, it forms local heat sources on specific surfaces (e.g., floors, walls) through direct beam spots, driving complex natural convection and radiative heat exchange. To capture the resulting **Thermal Lag** and **Local Discomfort**, this section establishes a dual-layer thermal model ranging from the Transient Thermo-Convection Field (TTCF) to lumped parameter networks.

3.2.1 TTCF Modeling

Treating the indoor space as a 3D domain $\Omega = [0, L_x] \times [0, L_y] \times [0, L_z]$, the evolution of the temperature field $T(\mathbf{x}, t)$ follows the law of conservation of energy (e.g. Bergman [8]):

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k_{\text{eff}} \nabla T) + Q_{\text{solar}}(\mathbf{x}, t) \tag{9}$$

where $Q_{\text{solar}}(\mathbf{x}, t)$ is the spatial redistribution of the window heat gain power $\dot{Q}_{\text{solar,win}}(t)$ calculated in Section 3.1. Specifically, the direct component is mapped as a moving light spot heat source projected onto the floor or internal walls, while the diffuse component is treated as a diffuse heat source near the window. This spatial mapping directly links the geometric output of the VSSI model, ensuring physical consistency between heat input and solar position.

To reflect natural convection effects without explicitly solving the Navier-Stokes equations, we introduce an effective thermal diffusivity k_{eff}. Under the *Boussinesq* approximation, when the Rayleigh number induced by window heating exceeds $Ra > 10^9$, turbulent convection significantly enhances heat mixing. Therefore, we adopt an enhanced diffusion coefficient in the air domain to phenomenologically characterize this convective stirring effect, while maintaining the intrinsic thermal conductivity of materials in solid thermal mass regions (walls, slabs).

The model closure relies on boundary conditions. We adopt Robin boundary conditions on the exterior surfaces of the envelope. Denoting the outward normal as n:

$$-k \frac{\partial T}{\partial n} = U_{\text{eff}} \left(T - T_{\text{env}}(t) \right) \tag{10}$$

where $T_{\text{env}}(t)$ corresponds to the hourly outdoor temperature T_{out} or ground temperature T_g described in Section 2.2. This set of equations constitutes the physical cornerstone for capturing indoor microclimate distribution, providing necessary field data support for the analysis of "Cold Radiant Asymmetry" in Section 4.3.4.

3.2.2 Reduced-Order RC / Thermal Mass Model

To balance computational efficiency with physical accuracy, this study adopts a dual-layer solution strategy: the TTCF model constructed in Section 3.2.1 is used for fine-grained validation (e.g., window-side thermal comfort assessment), while a lightweight and efficient calculation kernel is required for parameter optimization involving thousands of iterations. For this purpose, we introduce the **Lumped Parameter Method** to construct an equivalent RC (resistance–capacitance) thermal network model.

The physical basis of this method lies in the natural duality between thermal and electrical systems: temperature difference ΔT drives heat flow $\dot{Q}$, just as voltage difference ΔV drives current I; thermal resistance R of materials impedes heat flow, while heat capacity C stores thermal energy, creating inertia.

Based on this, we treat the indoor air as a uniform temperature node $T_z(t)$ and construct an equivalent RC thermal network. The energy balance equation can be written directly by analogy to Kirchhoff's Current Law (KCL):

$$C_z \frac{dT_z}{dt} = \frac{T_{\text{out}} - T_z}{R_{\text{eq}}} + \dot{Q}_{\text{solar,win}}(t) + \dot{Q}_{\text{int}}(t) + \dot{Q}_{\text{HVAC}}(t) \tag{11}$$

where $R_{\text{eq}} = 1/(UA)$ is the total thermal resistance of the envelope. C_z is the equivalent heat capacity of the room. To reflect the thermal inertia of furniture and internal partitions without explicitly modeling them, we adopt the method of **equivalent specific heat conversion**:

$$c_{p,\text{eff}} = c_{p,\text{air}} + \frac{C_{\text{wall}} + C_{\text{furniture}}}{\rho_{\text{air}} V_{\text{air}}} \tag{12}$$

This Ordinary Differential Equation (ODE) clearly reveals the essence of transient response: the system output $\dot{Q}_{\text{HVAC}}$ must not only offset current heat transmission losses and radiative gains but also counteract the rate of temperature change determined by C_z, which is the mathematical root of the "Thermal Lag" phenomenon.

To more precisely capture the thermal inertia of the envelope (e.g., heavy brick walls), we refine the wall into a ***3R2C*** network (Figure 5). In this second-order model, the wall is split into an outer surface node T_{wo} and an inner surface node T_{wi}, whose dynamic process is described by the following system of differential equations:

$$\begin{cases} C_{\text{out}} \frac{dT_{wo}}{dt} = \frac{T_{\text{out}} - T_{wo}}{R_{\text{out}}} - \frac{T_{wo} - T_{wi}}{R_{\text{wall}}} + \alpha I_{\text{sol}}, \\ C_{\text{in}} \frac{dT_{wi}}{dt} = \frac{T_{wo} - T_{wi}}{R_{\text{wall}}} - \frac{T_{wi} - T_z}{R_{\text{in}}}. \end{cases} \tag{13}$$

Here, solar radiation αI_{sol} is first absorbed by the outer node and must overcome the intermediate thermal resistance R_{wall} and "charge" the capacitors $C_{\text{in/out}}$ before affecting the interior. This mathematical structure successfully simulates the amplitude attenuation and phase lag characteristics of heavy walls in reality.

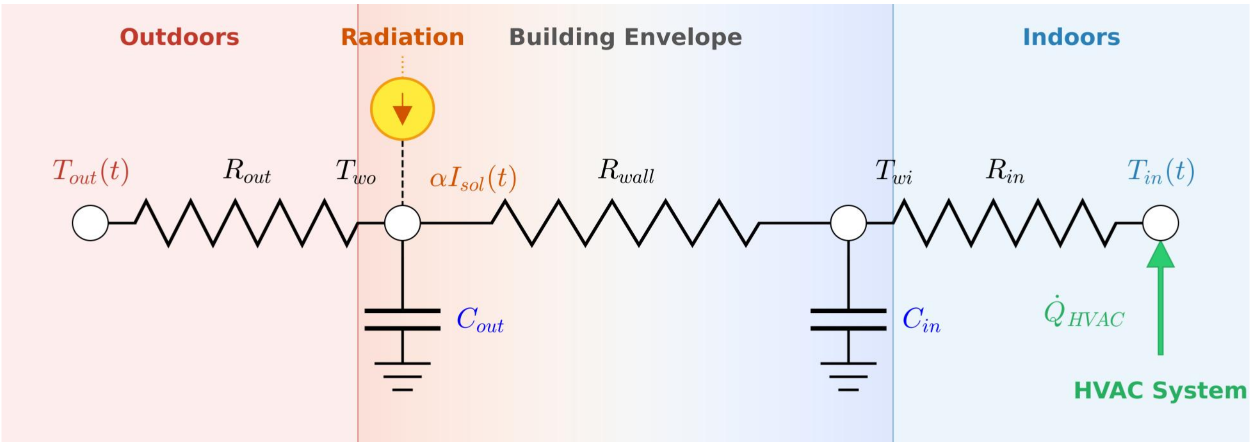

Figure 5: 3R2C equivalent thermal circuit; envelope as low-pass filter for diurnal perturbations.

The hourly dynamic load sequence $\dot{Q}_{\text{HVAC}}(t)$ calculated based on these equations directly constitutes the core objective function (energy consumption/cost) for parameter optimization in Chapter 4. This enables us to traverse and screen tens of thousands of design combinations at extremely low computational cost.

3.2.3 Thermal Lag and Spatial Temperature Field

Although the RC model macroscopically quantifies the system's thermal inertia, to deeply reveal "why the peak load is delayed by several hours," we need to return to the analytical level of continuous media. When solar radiation acts on the outer surface of the envelope with a diurnal period $\omega = 2\pi/24\text{h}$, the propagation of the thermal wave inside the wall exhibits classic **Decrement-Delay** characteristics.

Solving the heat diffusion equation for a 1D homogeneous wall $\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$ in the frequency domain yields the complex analytical solution for the thermal wave at depth x:

$$T(x,t) = \Re\left\{\Delta T_{\text{surf}} \cdot e^{-x/\mu} \cdot e^{j(\omega t - x/\mu)}\right\}, \quad \text{with} \quad \mu = \sqrt{\frac{2\alpha}{\omega}} \tag{14}$$

In the above equation, the phase term $-x/\mu$ directly reveals the physical root of the time lag. For a wall of thickness L, its heat transfer time lag t_{lag} can be explicitly expressed as:

$$t_{\text{lag}} = \frac{L}{\mu \cdot \omega} = L\sqrt{\frac{1}{2\alpha\omega}} \tag{15}$$

This formula provides solid mathematical support for the high agreement between the simulation results (approx. 3.0 h) and the theoretical value (2.87 h) in Section 4.2.3, proving the correctness of our model in capturing dynamic thermal processes.

Furthermore, to compensate for the RC model's lack of spatial resolution, the TTCF model supplements 3D temperature field information. By solving the convection-diffusion equation, it precisely characterizes local hotspots and vertical thermal stratification induced by non-uniform window-side heating, ensuring the final solution meets rigorous standards for local thermal comfort in "space" and peak shifting in "time."

3.3 CASTO Framework

The VSSI geometric model and the TTCF/RC thermal response models described above constitute the simulation kernel for building thermal performance. To find the optimal solution for specific climates within a multidimensional design space, this section encapsulates it into a **Climate-Adaptive Shading-Thermal Optimization (*CASTO*)** mathematical problem. This framework achieves automatic optimization from design parameters to performance indicators through a closed-loop feedback chain of "Geometry - Energy - Thermodynamics."

3.3.1 Mathematical Formulation of the Optimization Problem

We formulate the design problem as a non-linear constrained multi-objective programming problem. The decision vector $\mathbf{x}$ contains key parameters in three dimensions: geometric, optical, and thermal:

$$\mathbf{x} = [L_{\text{overhang}},\ L_{\text{fin}},\ \tau_{\text{eff}},\ C_{\text{eff}}]^T \tag{16}$$

where L_{overhang} and L_{fin} are the horizontal and vertical shading depths (geometric dimension), τ_{eff} is the effective transmittance of the window system (optical dimension, corresponding to SHGC), and C_{eff} is the thermal mass enhancement multiplier relative to the baseline building (thermal dimension).

To comprehensively evaluate building performance, we constructed a triangular evaluation system including thermal comfort, luminous environment, and economic efficiency as the objective functions and constraints for optimization.

(1) Thermal Comfort: Operative Temperature and Comfort Rate

Traditional energy simulation often focuses only on air temperature T_a, but in high-performance buildings, the influence of the inner surface temperature T_{si} of the envelope on human thermal sensation cannot be ignored. Therefore, we adopt the operative temperature T_{op} as the core indicator:

$$T_{op} = \frac{T_a + T_{\text{mrt}}}{2}, \qquad T_{\text{mrt}} = \sum_i F_{p\text{-}i}\, T_{si} \tag{17}$$

where T_{mrt} is the mean radiant temperature. Annual comfort is characterized by the comfort coverage rate η_{comfort}, which is the percentage of hours in the year when T_{op} falls within the comfort zone (e.g., $20 \sim 26°\text{C}$).

(2) Visual Comfort: Daylight Proxy Model

While excessive shading reduces cooling loads, it can lead to dim interiors; insufficient shading causes glare. We use a daylight proxy metric DF_{proxy} to quantify this trade-off:

$$\text{DF}_{\text{proxy}} = \tau_{\text{eff}} \cdot (1 - \overline{F_s}) \cdot \frac{A_{\text{win}}}{A_{\text{floor}}} \tag{18}$$

This metric is positively correlated with Spatial Daylight Autonomy (sDA) and serves as a hard constraint for filtering out "darkroom effect" solutions (requiring sDA $\geq 50\%$).

(3) Economic Efficiency: Annual Cumulative Thermal Load

Regardless of whether the climate is cooling-dominated or heating-dominated, the core optimization goal is to minimize the mechanical energy injection required to maintain room temperature. We define the annual cumulative energy load J_{load} as:

$$J_{\text{load}}(\mathbf{x}) = \int_{\mathcal{T}} \left| \dot{Q}_{\text{HVAC}}(\mathbf{x}, t) \right| \, dt \tag{19}$$

where $\dot{Q}_{\text{HVAC}}$ is the instantaneous ideal heating, ventilation and air conditioning (HVAC) load output by the RC model (positive for heating, negative for cooling). For single-condition regions (like Sungrove), this integral corresponds to cooling energy; for mixed conditions, it represents the total annual thermal regulation energy consumption.

Integrating the above indicators, the CASTO optimization problem is formulated as finding a set of Pareto optimal solutions that balance energy consumption, cost, and discomfort, subject to visual comfort constraints:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{J}(\mathbf{x}) = \left[\underbrace{J_{\text{load}}(\mathbf{x})}_{\text{Energy}}, \ \underbrace{1 - \eta_{\text{comfort}}(\mathbf{x})}_{\text{Discomfort}} \right] \\ \text{s.t.} \quad & \mathrm{DF}_{\text{proxy}}(\mathbf{x}) \geq \gamma_{\text{light}} \end{aligned} \tag{20}$$

This model achieves a closed-loop mapping from design parameters $\mathbf{x}$ to performance indicators $\mathbf{J}$ by coupling the radiation calculation in Section 3.1 with the thermal response simulation in Section 3.2.

3.3.2 Solution: NSGA-II and TOPSIS

Faced with the conflicting objectives mentioned above (e.g., reducing energy consumption often means sacrificing daylighting), a single optimal solution does not exist. We formalize the optimization problem as finding the Pareto front in the decision space $\mathbf{x}$. Under the NSGA-II (Non-dominated Sorting Genetic Algorithm II) framework, we first rapidly scan thousands of parameter combinations through the RC network model to filter out non-dominated solutions; subsequently, we use *TOPSIS* (Technique for Order Preference by Similarity to Ideal Solution) to comprehensively consider energy, comfort, and glare risk, selecting the recommended solution $\mathbf{x}^*$ with the most engineering value from the Pareto front.

Algorithm 1 NSGA-II Annual Multi-Objective Optimization + TOPSIS Decision

Input: Decision variables $\mathbf{x}$, hourly weather data, building parameters.
Output: Pareto front solution set and recommended solution $\mathbf{x}^*$.
Initialization: Population $N = 16$, Generations $G = 20$; Select four representative days.
Simulation Loop: For each individual $\mathbf{x}_k$:
  Calculate Shading: $F_s(t) \leftarrow \mathcal{G}(\vec{S}(t), L_{\text{overhang}}, L_{\text{fin}})$
  Calculate Heat Gain: $\dot{Q}_{\text{solar}}(t) \leftarrow F_s(t) \cdot \tau_{\text{eff}} \cdot I(t) \cdot A_{\text{win}}$
  Solve RC Network: Get $T_{\text{in}}(t)$, calculate HDH, CDH, sDA.
Constraint Check: If sDA $< 30\%$, mark as infeasible.
Evolution: Non-dominated sorting, crowding distance, selection, crossover, mutation.
Decision: Normalize Pareto set by (HDH, CDH, Glare), TOPSIS to rank and output $\mathbf{x}^*$.

4 Case Study

4.1 Case 1: Sungrove University (Cooling-Dominated)

The climate boundary for Sungrove is derived from hourly weather data for Honolulu, USA (21.3°N). Located south of the Tropic of Cancer, this site features a typical tropical semi-arid climate (BSh/As). The model utilizes its consistently high solar altitude throughout the year and sustained background cooling load to construct a purely "Cooling-Absolutely-Dominated" scenario, forcing the design to maximize the geometric reduction of direct beam spots. Table 3 presents the solar angles and thermal environment boundaries for Sungrove near noon on representative days of the four seasons.

Table 3: Solar angles and thermal boundaries for typical solar terms in Sungrove (near noon). Note: Azimuth is 0° for North, clockwise.

Term	Date	δ (°)	ω (°)	α (°)	γ_s (°)	θ_i (°)	$T_{\min}$	$T_{\max}$	ΔT_{out}	T_g
Spring	03-20	-0.46	2.00	69.96	180.43	69.96	19.0	28.0	9.0	23.0
Summer	06-21	23.45	0.50	86.56	359.22	93.44	28.0	38.0	10.0	28.0
Autumn	09-22	0.64	-1.75	70.20	180.22	70.20	24.0	33.0	9.0	27.0
Winter	12-21	-23.42	-0.50	46.58	179.98	46.58	15.0	24.0	9.0	22.0

4.1.1 Baseline Climate and Thermal Behavior

The core pain point of Sungrove is "Heat Surplus." On the Summer Solstice, the solar altitude angle at noon approaches 90°, creating a "shadowless" phenomenon; even on the Winter Solstice, the altitude remains above 45°, implying strong radiation input year-round.

Under the baseline scenario without any passive strategies (double-pane standard glass, no shading), simulations show significant overheating indoors on the Summer Solstice:

1. **Extreme Indoor Temperature Rise:** Under a peak outdoor temperature of 38°C, the indoor peak temperature soars to 48.4°C, forming a typical "Greenhouse Effect."
2. **Severe Spatial Non-uniformity:** Areas near windows are exposed to direct sunlight, with a temperature difference of up to 22°C compared to the depths of the room, severely destroying spatial usability.

This baseline analysis indicates that relying solely on mechanical cooling is not only energy-intensive but also unable to resolve thermal discomfort caused by radiation.

4.1.2 Climate-Adaptive Optimization and Performance

Applying the *NSGA-II* framework (Section 3.3), we ran an annual search in the 4D space (L_{overhang}, L_{fin}, τ_{eff}, C_{eff}) and converged to:

$$L^*_{\text{overhang}} = 1.0\,\text{m}, \quad L^*_{\text{fin}} = 0.8\,\text{m}, \quad \tau^*_{\text{eff}} = 0.40, \quad C^*_{\text{eff}} = 1.5\times \tag{21}$$

This set defines the retrofit strategy. The geometric operator $\mathcal{G}$ is realized as a 1.0 m overhang (intercepting high-altitude noon sun, $F_s \approx 0.54$) plus 0.8 m vertical fins (blocking low-angle morning/evening sun), reducing east–west daily heat gain from 120 kWh to 35 kWh. The material operator $\mathcal{M}$ with $\tau_{\text{eff}} = 0.40$ filters 60% of transmitted radiation at the source.

Figure 6 quantifies the improvement. The load curve (a) flattens: daily solar gain falls from 576 kWh to 212 kWh (**63.1%** reduction). The temperature histogram (c) shifts left—mean indoor temperature drops from 36.1°C to 26.5°C, eliminating extreme overheating risk.

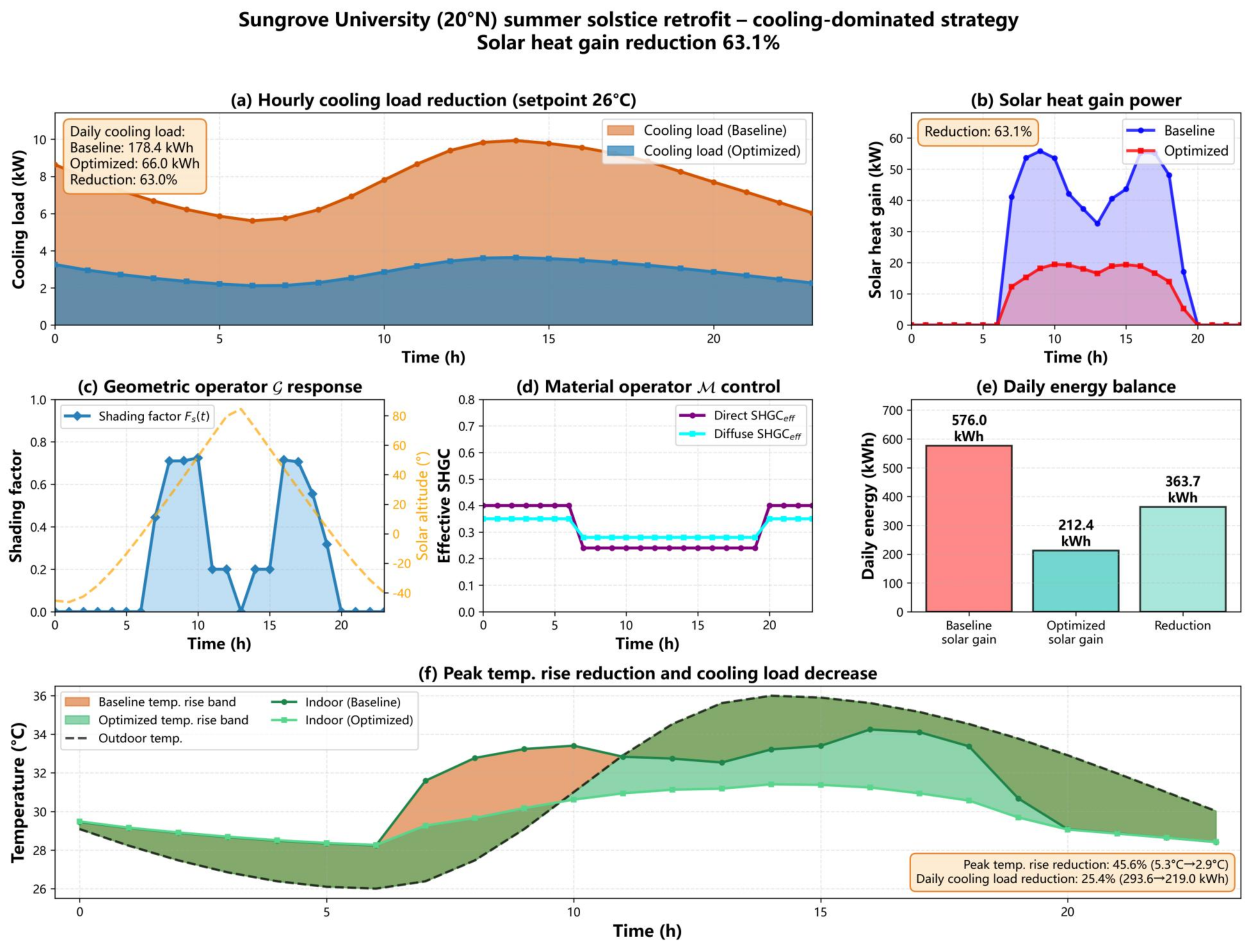

Figure 6: Comprehensive analysis of cooling load, solar heat gain, and shading effects for Sungrove on Summer Solstice.

Pareto front analysis (Figure 6d) shows that beyond $L = 1.2$ m energy gains diminish while sDA falls below 50% ("darkroom effect"). The chosen $L = 1.0$ m sits at the elbow of the energy–daylighting trade-off, giving strong heat rejection with sDA $\approx$ 96%.

Figure 7 compares thermal fields at 17:00 (Summer Solstice). In the elevation view (left), the baseline (top) shows oblique light driving window-side temperature to 35.0°C and strong vertical stratification; the optimized case (bottom) cuts spot intensity 70% via overhangs, fins, and louvers, holding the second floor near 33.8°C. The plan view (right) shows spot projection and uniformity: the optimized layout pushes spots inward and removes the window-side hot zone. Together with the 63.1% load reduction, this gives numerical–visual confirmation of the shading strategy.

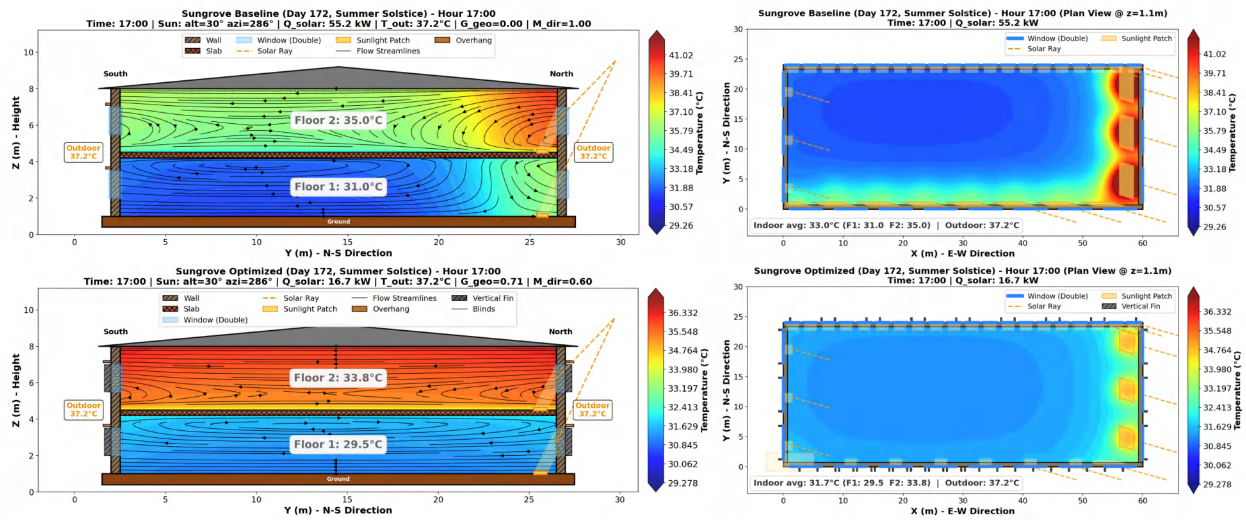

Figure 7: Comparison of indoor thermal fields between Baseline and Optimized scenarios for Sungrove (Summer Solstice 17:00). Top row: Baseline (no shading); Bottom row: Optimized (Overhang + Fins + Louvers). Left: N-S section view showing solar ray trajectories and vertical stratification; Right: Plan view showing horizontal spot projection and temperature uniformity. The optimized scheme significantly reduces window-side overheating via geometric blocking, lowering indoor temp from 31.0°C to 29.5°C.

4.1.3 Thermal Lag and Energy Self-Consistency

Shading reduces the "amplitude" of energy input at the source, while thermal mass and active strategies determine the distribution and balance of energy on the time axis. This section reveals the complete physical logic from "Load Shifting" to "Energy Self-Consistency."

(1) Thermal Lag & Peak Shifting

Solar radiation peaks at noon, but the peak indoor cooling load is delayed to 14:00–16:00; the simulated lag (**3.0 h**) matches the analytical $t_{\mathrm{lag}} = (L/\mu)/\Omega_{24}$ (2.87 h, Section 3.2.4), shifting demand to the afternoon and aligning with peak-valley pricing. To counter night-time temperature rise from daytime heat storage, **Night Ventilation** (high-side windows 22:00–6:00) uses cool night air to reset slab temperature to ~ 26°C by dawn, reserving thermal capacity for the next day. Figure 8 illustrates the resulting net-zero cooling pathway and energy self-sufficiency.

4.2 Case 2: Borealis University (Heating-Dominated)

For the high-latitude location of Borealis University, we introduced *TMY3* data from Fairbanks, USA (64.8°N) to simulate the extreme thermal environment of a Subarctic Continental Climate (Dfc). This dataset embodies the dramatic contrast between "Winter

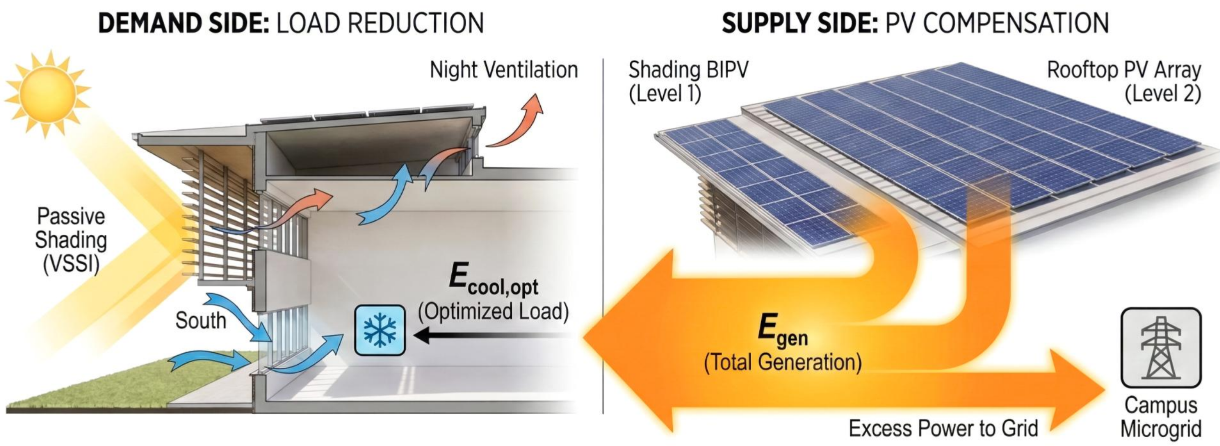

Figure 8: Net-zero cooling pathway.

Solstice Polar Night dim light" and "Summer Solstice Polar Day heat accumulation," making it an ideal sample for testing the "Peak Shifting and Valley Filling" capability of Thermal Mass strategies in deep cold environments.

Table 4 presents the solar angles and thermal environment boundaries for Borealis near noon on representative days of the four seasons.

Table 4: Solar angles and thermal boundaries for typical solar terms in Borealis (near noon). Note: Azimuth is 0° for North, clockwise.

Term	Date	δ (°)	ω (°)	α (°)	γ_s (°)	θ_i (°)	$T_{\min}$	$T_{\max}$	ΔT_{out}	T_g
Spring	03-20	-0.46	2.00	24.99	180.16	24.99	-14.0	-4.0	10.0	0.0
Summer	06-21	23.45	0.50	48.45	180.07	48.45	13.0	27.0	14.0	10.0
Autumn	09-22	0.64	-2.00	25.23	179.80	25.23	2.0	10.0	8.0	8.0
Winter	12-21	-23.42	-0.50	1.87	179.99	1.87	-22.0	-14.0	8.0	2.0

4.2.1 Challenges in Extreme Cold

The core contradiction of Borealis lies in the extreme spatiotemporal mismatch between thermal supply and demand. On the Winter Solstice, the solar altitude at noon is only 1.87°, the sunshine window is compressed to hours (9:00–15:00), and peak irradiance is low (about 300 W/m^2). Meanwhile, outdoor temperatures remain below -20°C for long periods. Under these conditions, traditional "Passive Defense" relying solely on insulation is insufficient. The system must shift to "Active Capture and Storage," maximizing solar heat gain within the limited window and shifting it to the night when demand peaks.

4.2.2 Capture-Store Coupling Mechanism

To solve the aforementioned spatiotemporal mismatch, we constructed a system solution coupling high-transmittance envelopes with phase-change thermal mass.

First, in the "Capture" phase, we replaced traditional double glazing with a **Triple-pane Argon-filled Glazing** system. After optimization, this system maintains an extremely low heat transfer coefficient ($U \approx 0.75\,\mathrm{W}/(\mathrm{m}^2 \cdot \mathrm{K})$) while selecting optical coatings with high SHGC = 0.65. This “High Transmission, Low Transfer” characteristic makes it an efficient radiation trap, allowing winter sunlight to enter in the short-wave band while effectively blocking long-wave indoor heat loss.

Second, in the "Storage" phase, to "lock" the transient daytime heat gain indoors, we introduced *Micro-encapsulated Phase Change Materials (PCM)* (Cabeza et al. [9]) on the inner wall surfaces. The PCM phase change temperature is set to $T_m \approx 22°\mathrm{C}$, located exactly at the center of the human thermal comfort zone. When the indoor temperature rises due to solar gain, the PCM undergoes a phase change to absorb heat, converting sensible heat into latent heat storage. Its equivalent heat capacity C_{eff} instantly increases several times:

$$C_{\text{eff}} = C_s + \frac{L_f}{\Delta T_{\text{trans}}} \tag{22}$$

This mechanism significantly increases the penetration depth of the thermal wave $\delta = \sqrt{2\alpha_{\text{eff}}/\omega}$, thereby substantially enhancing the system’s thermal inertia without increasing the structural thickness of the building.

4.2.3 Transient to Annual Performance

Figure 9 quantifies the efficacy of these strategies. Heat flow (c) shows the optimized scheme widening the energy inlet via high-SHGC glass, with total solar gain +38% vs. baseline. The transient response (a) is more telling: the baseline (orange) exhibits fast rise, fast fall” with temperature dropping below comfort after sunset and **high-load night heating**; the optimized case (blue) gains 2.8 h thermal lag from *PCM* latent heat, smoothing daytime peak release into the night. The temperature histogram (b) confirms this peak shaving and valley filling”—optimized distribution converges to the comfort zone, reducing heating start-stops and cutting night peak power by nearly 40%.

Figure 10 compares thermal fields at 13:00 (Spring Equinox). In the elevation view (left), the baseline (top) has weak envelope storage and second-floor temperature 13.1°C; the optimized case (bottom) uses triple argon glass and wall thermal mass, raising that to 14.9°C. The plan view (right) shows spot projection and uniformity—optimized spots are more concentrated and indoor temperature level is raised. Together with +38.3% solar gain and reduced temp. swing (7.6 to 4.8), this corroborates the high-transmission glass + *PCM* thermal mass strategy for heating-dominated climates.

4.2.4 Radiant Asymmetry and Thermal Comfort

In the extreme cold climate of Borealis, Air Temperature is insufficient to fully describe human thermal sensation; **Cold Radiant Asymmetry** becomes the key short board restricting comfort.

Baseline scenario simulations show that when the outdoor temperature drops to −10°C, the inner surface temperature of ordinary double glazing is only about 12°C. At this time,

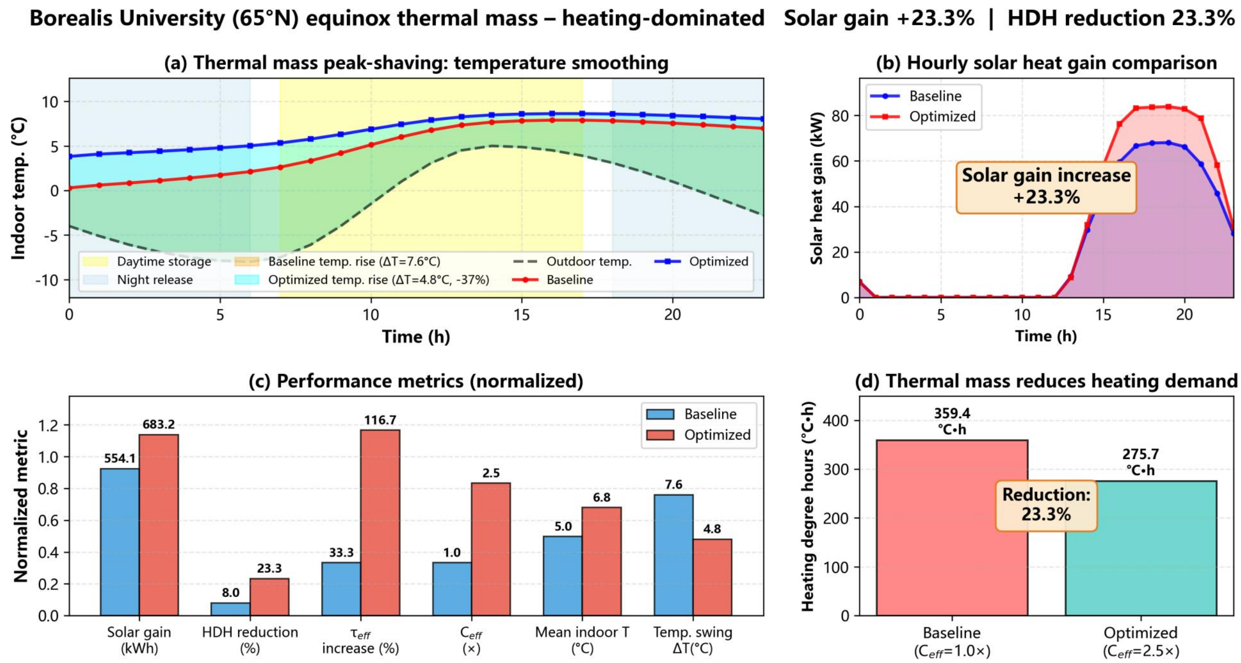

Figure 9: Borealis thermal mass strategies (Winter Solstice): heat storage/release.

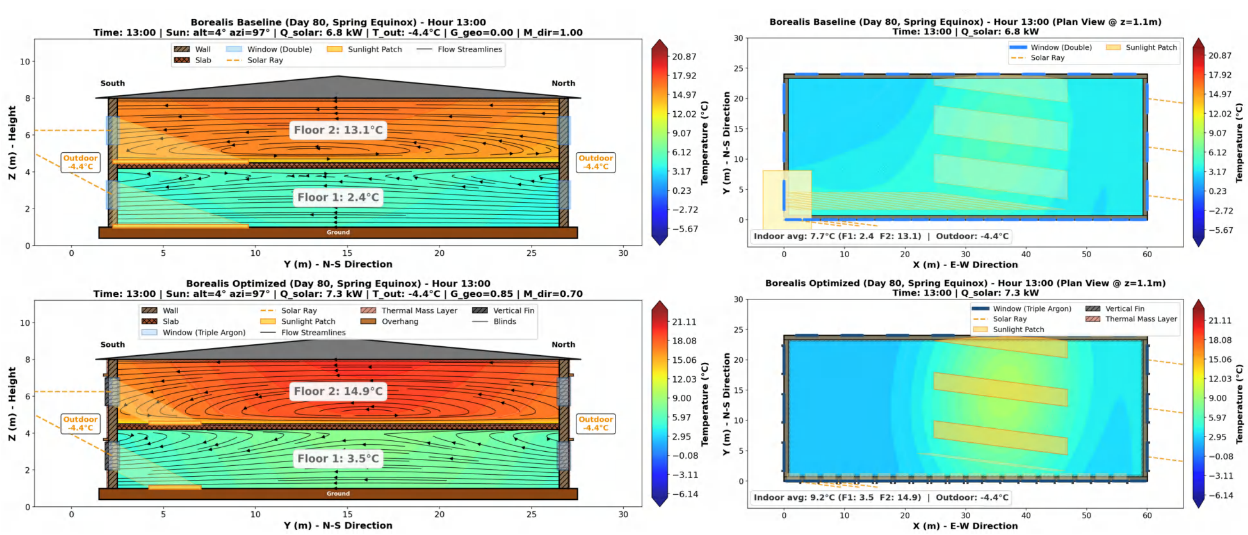

Figure 10: Baseline vs optimized thermal fields, Borealis (Spring Equinox 13:00).

even if the HVAC system forcibly heats the air to 22°C, people near the window will still feel cold due to strong radiative heat loss to the cold glass surface (i.e., $T_{\mathrm{mrt}} \ll T_a$).

The triple glazing and high heat capacity walls in the optimized scheme fundamentally correct this radiation field. As shown in the thermal field simulation in **Figure 10**, the inner surface temperature of the optimized envelope has generally increased by more than 4°C, significantly raising the Mean Radiant Temperature T_{mrt}. This means that at the same air temperature, the human Operative Temperature T_{op} is higher, and the comfort zone is actually widened. Only the TTCF field model can support this local-comfort verification; the RC model does not resolve spatial T_{mrt}.

4.3 New Student Union Building

Figure 11 presents our Student Union prototype as an *envelope-first, climate-robust system*: a PV-ready roof (energy offset), a glazed atrium spine (stack purge), and a module-based adaptive screen shifting between day curled/closed and night open/unfurled states. This targets shading that remains effective under future climate conditions.

4.3.1 Concept and Geometric Parameterization

To keep the prototype computable within our $VSSI \rightarrow F_s(t) \rightarrow$ heat-gain pipeline, we parameterize the envelope with four variables: overhang depth L, glazing SHGC, effective thermal mass C_{th}, and window-to-wall ratio (WWR). The adaptive screen state is an opening ratio $r(t) \in [0, 1]$ (0 closed, 1 open), mapped to the time-resolved shading factor:

$$F_s(t) = \mathrm{clip}\Big(\eta[1 - r(t)], 0, 1\Big), \tag{23}$$

where η is the effective coverage factor of the module array. Physically, the screen is a **Passive Adaptive Skin** of ***thermo-bimetal*** triangular modules arranged in a *Penrose tiling* pattern for aperiodic coverage. When temperature rises, modules curl to increase shading and open ventilation paths; when it drops, they flatten to increase light transmission and heat gain—a geometric carrier of the "Shading–Ventilation" coupling.

4.3.2 Data-Driven Tuning and Climate-Robust Performance

We rank *Pareto* candidates using *TOPSIS*. The top designs at Polar (65°) cluster into a stable range: SHGC ≈ 0.66–0.68, WWR ≈ 0.22–0.24, and $C_{\mathrm{th}} \approx 185$–$204\,\mathrm{MJ/K}$, indicating that robust performance results from combining moderate-to-high solar admission with sufficient thermal inertia. Across the best, energy intensity is low (EUI ≈ 35–43 $\mathrm{kWh/m^2{\cdot}yr}$) and payback is short, while daylight autonomy is the bottleneck (sDA ≈ 18–24%).

Future-climate robustness (stress-test). Over a 30-year horizon (e.g., to mid-century), we evaluate the same metrics under boundary perturbations:

$$T_{\mathrm{out}}^{(k)}(t) = T_{\mathrm{out}}(t) + \Delta T_k, \qquad I_{\mathrm{beam}}^{(k)}(t) = (1 + \Delta I_k)\, I_{\mathrm{beam}}(t), \tag{24}$$

and select the tuning that minimizes worst-case energy penalty subject to a daylight constraint $P_{\mathrm{day}} \geq P_{\mathrm{min}}$. Table 5 lists the top five envelope candidates for the Polar (65°) case.

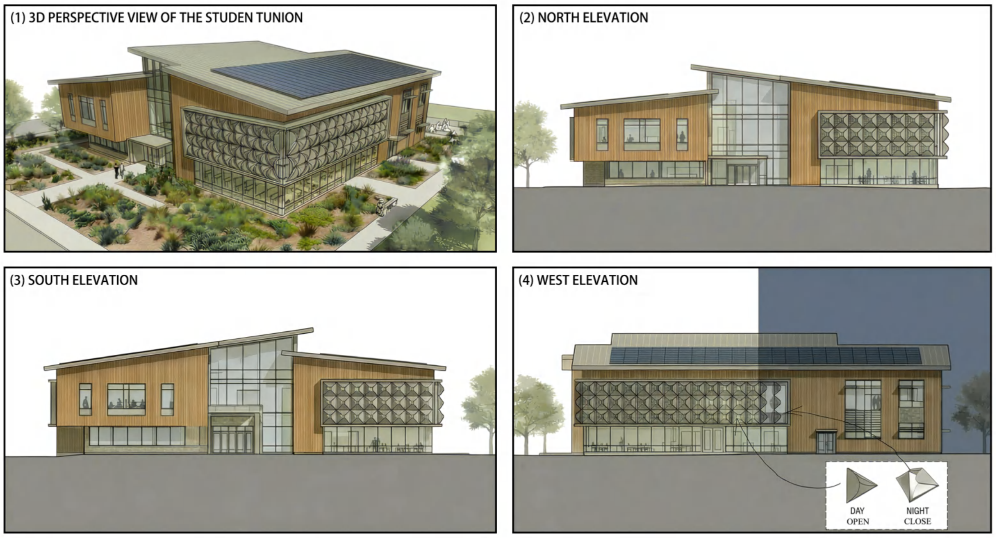

Figure 11: Student Union prototype: envelope-first, climate-robust system with PV roof, atrium spine, and thermo-bimetal adaptive screen (day curled / night open).

Table 5: Top 5 envelope candidates, Polar 65°.

Rank	L (m)	SHGC	C_{th} (MJ/K)	WWR	TOPSIS	EUI (kWh/m^2·yr)	HVAC (kWh/yr)	CO_2 (kg/m^2·yr)	sDA (%)	30y LCC ($)	Payback (yr)
1	1.22	0.679	203.8	0.24	0.8796	34.94	100619	15.73	19.80	415121	3.72
2	1.54	0.659	185.2	0.23	0.8725	35.10	101089	15.80	17.98	412208	3.64
3	0.57	0.669	92.4	0.22	0.8474	40.32	116129	18.15	18.75	436695	3.55
4	0.92	0.685	208.8	0.29	0.8411	36.47	105043	16.42	24.22	459945	4.49
5	0.73	0.587	175.8	0.20	0.8245	42.86	123444	19.29	18.91	442785	3.37

4.3.3 Shading Strategy and Dual-Climate Behavior

In the **Sungrove** scenario, the adaptive screen is oriented toward heatwaves and intense solar radiation: by day it curls to increase shading ($F_s \uparrow$) and reduce direct input; at night it opens to promote purge. In the **Borealis** scenario, the screen can serve as the outer layer of a double-skin facade—a greenhouse buffer in winter to reduce transmission loss and chimney-effect ventilation in summer to suppress overheating. The $r(t)$ and $F_s(t)$ rules are fed back into the heat-exchange boundary conditions in Section 3 to evaluate annual performance under different climates and future perturbations.

5 Sensitivity and Transferability

5.1 Setup and Global Structure

Sensitivity was evaluated with respect to the composite metric J_{comp} (thermal load, comfort deviation, daylight feasibility) using the same solar–thermal chain as in Chapter 3. A normalized OAT (one-at-a-time) index was adopted: $\Sigma_p = (p/J_{\text{comp}})\, \partial J_{\text{comp}}/\partial p$, with perturbations around the optimized baseline and all comfort/daylight constraints enforced.

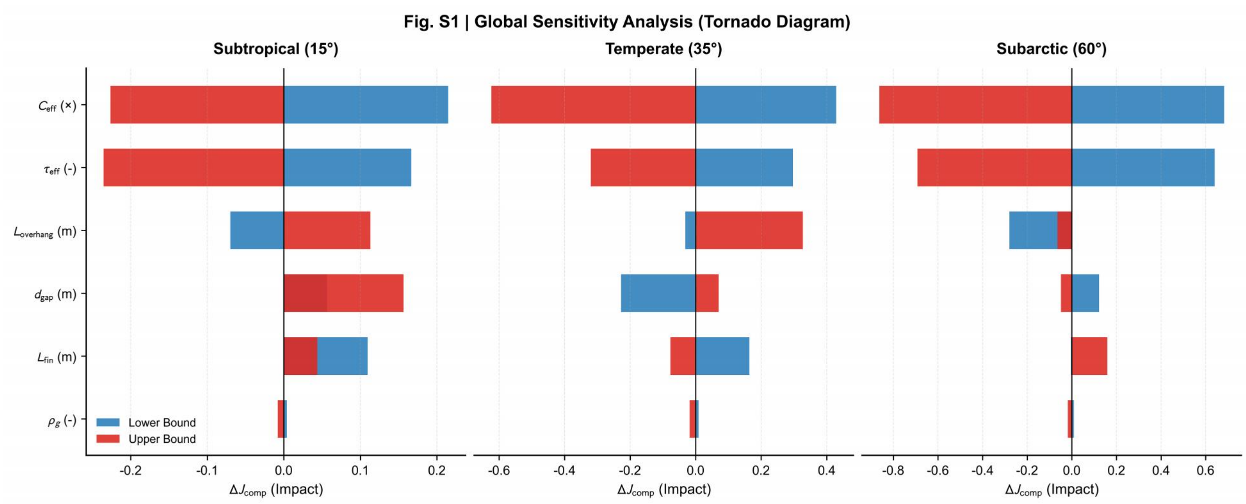

Figure 12: Global sensitivity (tornado diagram). Impact ΔJ_{comp} of six design parameters across Subtropical (15°), Temperate (35°), and Subarctic (60°).

Figure 12 shows that in cooling-dominated, low-latitude conditions the performance is most sensitive to τ_{eff} and L_{overhang}; as latitude increases, sensitivity shifts toward thermal mass C_{eff}, consistent with temporal heat storage under short winter daylight. The relative ordering of decision variables is stable within each regime and follows physical expectations.

5.2 Transferability and Implications

Figure 13 maps normalized sensitivities on a latitude–diurnal time plane: geometric shading parameters dominate during high solar altitude, while τ_{eff} remains broadly uniform; interaction between L_{overhang} and τ_{eff} is synergistic. Sensitivity patterns are continuous and interpretable across the domain, indicating climate-driven rather than case-dependent mechanisms. The framework is thus robust and transferable; design priorities shift predictably with latitude and solar geometry, and remain consistent with Section 4 results.

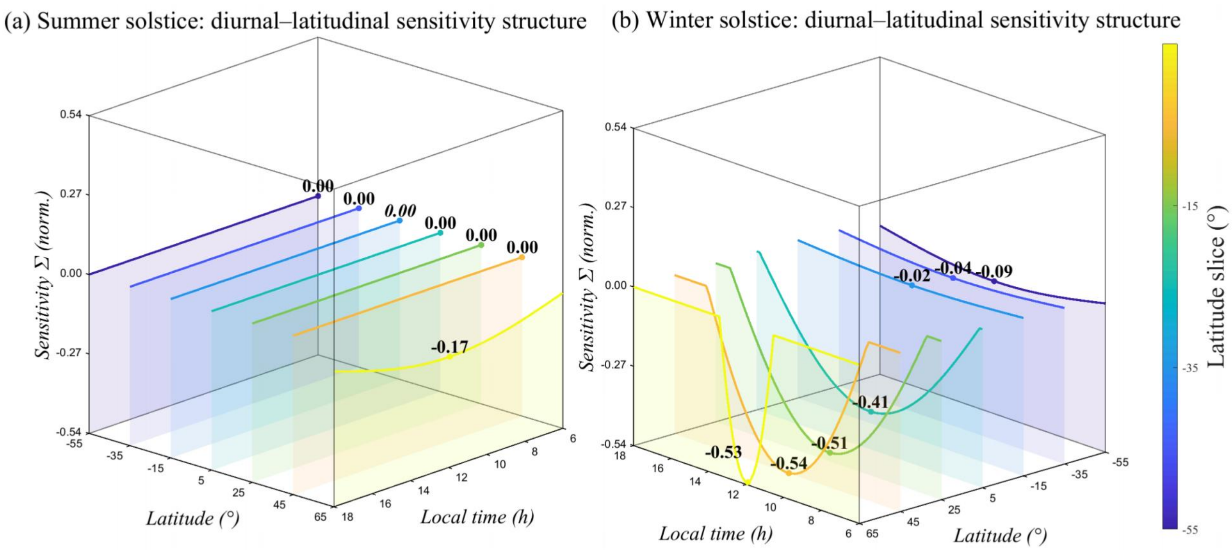

Figure 13: Diurnal–latitudinal sensitivity structure. (a) Summer solstice; (b) Winter solstice. Normalized sensitivity Σ vs. latitude and local time; L_{overhang}–τ_{eff} interaction.

6 Model Evaluation

6.1 Strengths: High-Fidelity Coupling & Transferability

The *VSSI* model goes beyond static solstice snapshots by combining the NREL solar-position algorithm with polygon clipping, achieving angular MAE below 0.35° and capturing off-peak effects such as early-morning glare that simplified methods often miss. Linking TTCF (spatial field) and RC networks (temporal lag) ties geometric shading to thermodynamic response and gives a clear physical basis for the observed 3 h thermal lag. A climate-dominance index σ extends these case-specific outcomes into a general picture: the same framework yields 63.1% load reduction at Sungrove and +38% solar capture at Borealis, showing that the model transfers well between cooling- and heating-dominated regimes.

6.2 Limitations: Physics vs. Efficiency

To keep *NSGA-II* runs tractable, indoor convection is represented by an effective diffusivity k_{eff} instead of full Navier–Stokes, so local overheating in stagnation zones may be underpredicted. The isotropic diffuse-sky assumption ignores circumsolar brightening and can overestimate shading effectiveness by on the order of 5–8% in hazy or overcast weather. Behavior is also assumed deterministic (e.g., night ventilation applied at fixed times); in practice, user actions and equipment delays introduce variability that the current equations do not capture.

6.3 Future Work: Towards Digital Twins

Next steps would be to calibrate the model with real-time sensor data and Bayesian inference on C_{eff}, moving toward a digital-twin role over the building lifecycle; to add a Perez all-weather sky model and spectral ray tracing for complex glazing; and to represent occupancy and window use with Markov-chain-based models, so that net-zero targets stay meaningful under real-world usage patterns.

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Team 2602872

To: Sungrove University
Subject: A Quantified and Scalable Pathway to Reduce Glare and Cooling Load while Preserving Daylight

Dear Campus Planning and Facilities Management Team,

Under intensifying solar exposure and more frequent heat waves, and in support of Sungrove University's goal of achieving net-zero cooling by 2040, the campus faces a clear decision challenge:

To address this, we developed a decision-ready passive shading framework that treats shading as a measurable and scalable design asset, and applied it to two prototypes: a retroft of Academic Hall In, especially on east and west façades. In addition, thermal inertia delays indoor temperature peaks by 2–4 hours, often shifting them into the afternoon when heat stress and cooling demand are highest.

Recommendations — Academic Hall North

Upgrade the south façade overhang to an effective depth of ~1.2 m, blocking high-altitude summer sun while admitting winter solar gains.

Add controllable vertical fins or side shading on east and west glazing to reduce low-angle glare and lateral heat gains.

Co-optimize glazing (SHGC and insulation) with shading depth, and use shading elements as PV-ready structures where feasible.

Scenario analysis indicates a ~10°C reduction in peak indoor temperature and a double-digit percentage reduction in annual cooling demand, with on-site PV capable of offsetting a meaningful share (≈20–30%) of the avoided cooling electricity.

Adopt a passive, self-adaptive façade using thermo-bimetal triangular modules, which curl under high temperatures to increase it to preserve daylight and winter gains. The system is evaluated using the same hourly performance metrics as the retrofit, enabling direct comparison.

Budget and Funding Considerations

To support implementation, we recommend framing shading retrofits at the unit façade or bay scale, enabling phased investment. Based on typical campus retrofit benchmarks, the proposed measus driven primarily by reduced cooling energy and partial PV generation. A pilot-first approach can limit initial expenditure to <10% of full-building retrofit cost, while providing measured data for refinement. Funding pathways may include staged capital allocation, energy performance contracting, or third-party PV participation.

Implementation Pathway

We recommend starting with a single monitored façade bay to calibrate model predictions using measured irradiance, indoor temperature, and energy data, before scaling the validated sp-cifications into a campus-wide guideline and monitoring framework.

We would welcome further discussion on pilot siting, sensor configuration, and an implementation timeline aligned with Sungrove's capital planning cycle.

Sincerely,
COMAP Team

Report on Use of AI

1. **Google Gemini Advanced (DeepResearch Mode)**
   Query: "Collect and summarize recent academic literature (2015-2025) regarding the evaluation criteria for assessing the construction quality of passive shading systems. Additionally, search for international industry standards, quotas, and budget guidelines (e.g., ISO, ASHRAE)."
   Output: The AI aggregated data from multiple sources and provided a structured list of references, including:

   - *ISO 9001:2015* for Quality Management Systems framework.
   - *FIDIC Conditions of Contract* for defining quality responsibilities.
   - *RICS NRM* for cost breakdown and measurement rules.
   - *UNIFORMAT II* for elemental cost classification of exterior enclosures.

   Usage: We used these references to build the evaluation index system in Section 5. All citations were verified manually.

2. **OpenAI ChatGPT (Model GPT-5.2 Thinking)**
   Query: "Could you provide specific examples of passive shading in real-world applications? Are there representative buildings in different geographical locations around the world that adapt to specific climates?"
   Output: Provided a list of representative buildings organized by climatic logic:

   - *Low-latitude (Cooling-dominated):* Buildings in Singapore employing deep overhangs and vertical fins.
   - *Temperate regions:* Fallingwater and European office buildings using fixed horizontal shading to admit winter sun.
   - *High-latitude (Heating-dominated):* Adjustable shading prioritizing winter solar gains while controlling glare.

   Usage: These examples were used to contextualize our problem background and validate the generalization of our model in Section 1.

3. **Anthropic Claude 4.5 Opus & OpenAI Codex**
   Query: "Please help me fix this Python code for the Transient Thermo-Convection Field (TTCF) model. The boundary condition loop is throwing an index error. Also, enhance the heatmap visualization to use a 'jet' colormap and include streamlines."
   Output: Identified the off-by-one error in the finite difference grid loop and provided a corrected script. It also generated a visualization function using `matplotlib` and `seaborn` to overlay velocity streamlines on the temperature field.
   Usage: The corrected code snippet was integrated into our solver. The visualization style was adopted for Figure 4 and Figure 6 in the paper.

4. **NanoBanana Pro (Generative Visualization Tool)**
   Query: "Generate a photorealistic 3D schematic diagram of a Biomimetic Adaptive Skin for a building facade. The skin should consist of triangular thermo-bimetal modules that curl open when heated and close when cooled. Architectural style: futuristic, sustainable."
   Output: Generated high-resolution concept images demonstrating the open/closed states of the adaptive skin.
   Usage: These images served as the visual basis for the "Student Union Prototype" design in Section 4.3. We manually annotated the generated images to explain the physical mechanism.

5. **DeepL Write & ChatGPT 4o**
   Query: "Please polish the following paragraph to be concise and academic: [Original Draft of Abstract]."
   Output: "[Polished Version] This paper proposes a Vector-based Solar-Shading Interaction (VSSI) model..."
   Usage: Used to refine the grammar and flow of the Abstract and Conclusion sections. All technical terminology was manually reviewed to ensure accuracy.

6. **OpenAI ChatGPT 4o (Data Analysis)**
   Query: "Write a Python script using Pandas to parse standard TMY3 (Typical Meteorological Year) weather data files (.epw format). I need to extract hourly 'Direct Normal Irradiance', 'Diffuse Horizontal Irradiance', and 'Dry Bulb Temperature' while skipping the header metadata rows."
   Output: Provided a robust Python function utilizing `pandas.read_csv` with correct delimiter settings and column indexing to extract the specific meteorological parameters required for Sungrove and Borealis locations.
   Usage: This script was used to preprocess the raw climate datasets for input into our Transient Thermo-Convection Field (TTCF) model. The extracted data integrity was verified by cross-referencing with official NOAA summaries.